

Microsoft - The Perfect Selfie

Team Name: MacroHard

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Project Objectives:

- **What it is:** The project is an integration of a drone, a video recording device, and a smartphone application that is capable of recognizing voice commands, using the drone to automatically orient a video recording device with respect to a subject, and uploading photos taken from the video device to a cloud storage media.
- **Current Video Recording Technology Limitations**
 - The majority of current video recording technologies has a limitation in that a person must physically move, and hold a video recording device in order to properly orient it with respect to the subject of the video.
 - The limitation is exacerbated when the photographer wants to be the subject of the photo.
 - **Current Drone Video Technologies:** follow me, Lily Camera, Iris+, Phantom
 - None of the current drone video technologies are capable of automatically orienting the video camera in front of the subject without the need for expensive additional equipment.
- **General Functionalities**
 - Smartphone application voice command to take details on activation and orientation of video recording device.
 - Smartphone application uses the phone's and drone's internal wifi and magnetometer to identify the physical orientation of the drone.
 - Smartphone application uses internal processor of the phone in order to calculate the drone commands needed to move drone to ideal orientation.
 - Drone uses wifi in order to receive necessary movement commands from smartphone application..
 - Drone sends video recording information (either a photograph or a video) to the smartphone through wifi..
 - Smartphone application is capable of automatically uploading video recordings and still images taken by drone to a cloud storage media.

Project Milestones

1. Set up Trello and GitHub - link them together

2. Researching technologies and agreeing on which technologies we will use for each feature of the project. Examples: which drone, what platform to write app, interfacing with cortana, way to communicate between device and drone.
3. Be able to use cortana to control device camera, uploading pictures, etc...
 - a. can have different camera modes
4. How to use device to communicate with drone:
 - a. app will translate voice commands to commands for drone
 - b. use wifi to send controls to drones
5. Tell Drone to take a picture
6. Use gps to have drone follow and take pictures, more camera modes
7. Have all features prototyped

Technologies

- Mobile App
 - Windows Phone 8 App created using C#/.NET framework
 - Using Cortana to interpret voice commands and translate them into commands for the drone to follow
 - Access camera and allow app to upload pictures to different social media sites/cloud
- Communication Technology
 - WiFi network between Drone and Smartphone
- Hardware
 - Parrot ARDrone 2.0
 - Features: GPS, WiFi connectivity, Dual cameras (one HD, one low res), magnetometer, 3-axis gyroscope, accelerometer, barometric sensor and ultrasound sensor.
 - Windows Phone

We will use the Agile process model. This allows us to work on a small portion of each feature at the same time. We will be able to constantly update and rework code and ideas. Constant communication within the team is paramount. Scrum meetings will be held daily as a way to check in with teammates and keep on track. The idea is to track progress and find any potential problem areas. Recognizing problem areas early allows the team to adapt and make changes before the problem gets out of hand. In addition we will use Trello in order to keep track of what objectives we are focusing on for each sprint. The idea is to have different list of "Things To Do," "In Progress," "Completed," and "Approved." As task is selected to be worked on it will be moved to the "In Progress" list. After it has been completed we will move it to the "Completed" list. After each item is completed, we will have a peer check over it or test it to make sure it is valid; it will then be moved to "Approved." The idea is that we are only working on one, maybe two, items at a time. Overall, our work flow is about communicating, being able to adapt and make continuous progress.

